

DETAILED JSON OUTPUT EXAMPLES

Complete Scenarios with Full Field Values

SCENARIO 1: Empty Frame (No People)

Input Parameters:

```
json

{
  "camid": 1,
  "org_id": 100,
  "userid": 42,
  "threshold": 10,
  "alert_rate": 80,
  "return_annotated": false,
  "confidence_threshold": 0.35
}
```

Output:

json

```
{  
  "camid": 1,  
  "org_id": 100,  
  "userid": 42,  
  "Frame_Id": "FR_1_1708268400000",  
  "Time_stamp": "2024-02-18T12:00:00.000000Z",  
  "Frame_Count": 1,  
  "Total_people_detected": 0,  
  "Current_occupancy": 0,  
  "People_ids": [],  
  "Entry_time": [],  
  "Exit_time": [],  
  "exitid": [],  
  "People_dwell_time": [],  
  "Confidence_scores": [],  
  "Bounding_boxes": [],  
  "x": [],  
  "y": [],  
  "w": [],  
  "h": [],  
  "accuracy": [],  
  "Total_entries": 0,  
  "Total_exits": 0,  
  "Net_count": 0,  
  "Occupancy_percentage": 0.0,  
  "Over_capacity_count": 0,  
  "Average_dwell_time": "00:00:00",  
  "Max_occupancy": 10,  
  "Status": ""
```

```
"is_alert_triggered": false,  
"Processing_Status": 1,  
"annotated_frame": null  
}
```

Key Points:

- All array fields are empty `[]`
 - Counters are 0
 - Status is empty string `""`
 - No alert triggered
-

SCENARIO 2: First Person Enters

Frame 10 Output:

json

```
{  
  "camid": 1,  
  "org_id": 100,  
  "userid": 42,  
  "Frame_Id": "FR_1_1708268410000",  
  "Time_stamp": "2024-02-18T12:00:10.000000Z",  
  "Frame_Count": 10,  
  "Total_people_detected": 1,  
  "Current_occupancy": 1,  
  "People_ids": ["P1_1"],  
  "Entry_time": ["2024-02-18T12:00:10.000000Z"],  
  "Exit_time": [],  
  "exitid": [],  
  "People_dwell_time": ["00:00:00"],  
  "Confidence_scores": [0.8934],  
  "Bounding_boxes": [[245, 180, 395, 480]],  
  "x": [320.0],  
  "y": [330.0],  
  "w": [150.0],  
  "h": [300.0],  
  "accuracy": [0.893],  
  "Total_entries": 1,  
  "Total_exits": 0,  
  "Net_count": 1,  
  "Occupancy_percentage": 10.0,  
  "Over_capacity_count": 0,  
  "Average_dwell_time": "00:00:00",  
  "Max_occupancy": 10,  
  "Status": ""
```

```
"is_alert_triggered": false,  
"Processing_Status": 1,  
"annotated_frame": null  
}
```

Key Changes from Empty Frame:

- `Total_people_detected`: $0 \rightarrow 1$
 - `Current_occupancy`: $0 \rightarrow 1$
 - `People_ids`: $[] \rightarrow ["P1_1"]$
 - `Entry_time`: $[] \rightarrow ["2024-02-18T12:00:10.000000Z"]$
 - `People_dwell_time`: $[] \rightarrow ["00:00:00"]$
 - All detection arrays populated with 1 element
 - `Total_entries`: $0 \rightarrow 1$
 - `Occupancy_percentage`: $0.0 \rightarrow 10.0$
-

SCENARIO 3: Person Still Present (5 seconds later)

Frame 15 Output:

json

```
{  
  "camid": 1,  
  "org_id": 100,  
  "userid": 42,  
  "Frame_Id": "FR_1_1708268415000",  
  "Time_stamp": "2024-02-18T12:00:15.000000Z",  
  "Frame_Count": 15,  
  "Total_people_detected": 1,  
  "Current_occupancy": 1,  
  "People_ids": ["P1_1"],  
  "Entry_time": ["2024-02-18T12:00:10.000000Z"],  
  "Exit_time": [],  
  "exitid": [],  
  "People_dwell_time": ["00:00:05"],  
  "Confidence_scores": [0.9123],  
  "Bounding_boxes": [[248, 182, 398, 482]],  
  "x": [323.0],  
  "y": [332.0],  
  "w": [150.0],  
  "h": [300.0],  
  "accuracy": [0.912],  
  "Total_entries": 1,  
  "Total_exits": 0,  
  "Net_count": 1,  
  "Occupancy_percentage": 10.0,  
  "Over_capacity_count": 0,  
  "Average_dwell_time": "00:00:05",  
  "Max_occupancy": 10,  
  "Status": ""
```

```
"is_alert_triggered": false,  
"Processing_Status": 1,  
"annotated_frame": null  
}
```

Key Changes:

- `People_dwell_time`: ["00:00:00"] → ["00:00:05"]
 - `Average_dwell_time`: "00:00:00" → "00:00:05"
 - `Bounding_boxes`, `x`, `y` slightly changed (person moved)
 - `Confidence_scores` updated
 - Same person ID "P1_1" tracked
-

SCENARIO 4: Person Exits (Critical Frame)

Frame 20 Output:

json

```
{  
  "camid": 1,  
  "org_id": 100,  
  "userid": 42,  
  "Frame_Id": "FR_1_1708268420000",  
  "Time_stamp": "2024-02-18T12:00:20.000000Z",  
  "Frame_Count": 20,  
  "Total_people_detected": 0,  
  "Current_occupancy": 0,  
  "People_ids": [],  
  "Entry_time": [],  
  "Exit_time": ["2024-02-18T12:00:20.000000Z"],  
  "exitid": ["P1_1"],  
  "People_dwell_time": [],  
  "Confidence_scores": [],  
  "Bounding_boxes": [],  
  "x": [],  
  "y": [],  
  "w": [],  
  "h": [],  
  "accuracy": [],  
  "Total_entries": 1,  
  "Total_exits": 1,  
  "Net_count": 0,  
  "Occupancy_percentage": 0.0,  
  "Over_capacity_count": 0,  
  "Average_dwell_time": "00:00:10",  
  "Max_occupancy": 10,  
  "Status": ""
```



```
"is_alert_triggered": false,  
"Processing_Status": 1,  
"annotated_frame": null  
}
```

Key Changes:

- `Total_people_detected`: 1 → 0
 - `Current_occupancy`: 1 → 0
 - `People_ids`: ["P1_1"] → []
 - `Entry_time`: ["..."] → []
 - `Exit_time`: [] → ["2024-02-18T12:00:20.000000Z"]  EXIT APPEARS
 - `exitid`: [] → ["P1_1"]  PERSON ID APPEARS
 - All detection arrays: cleared to []
 - `Total_exits`: 0 → 1
 - `Average_dwell_time`: "00:00:10" (from top 10 dwell times)
-

SCENARIO 5: After Exit (Next Frame)

Frame 21 Output:

json

```
{  
  "camid": 1,  
  "org_id": 100,  
  "userid": 42,  
  "Frame_Id": "FR_1_1708268421000",  
  "Time_stamp": "2024-02-18T12:00:21.000000Z",  
  "Frame_Count": 21,  
  "Total_people_detected": 0,  
  "Current_occupancy": 0,  
  "People_ids": [],  
  "Entry_time": [],  
  "Exit_time": [],  
  "exitid": [],  
  "People_dwell_time": [],  
  "Confidence_scores": [],  
  "Bounding_boxes": [],  
  "x": [],  
  "y": [],  
  "w": [],  
  "h": [],  
  "accuracy": [],  
  "Total_entries": 1,  
  "Total_exits": 1,  
  "Net_count": 0,  
  "Occupancy_percentage": 0.0,  
  "Over_capacity_count": 0,  
  "Average_dwell_time": "00:00:10",  
  "Max_occupancy": 10,  
  "Status": ""
```

```
"is_alert_triggered": false,  
"Processing_Status": 1,  
"annotated_frame": null  
}
```

Key Point:

- `Exit_time`: ["2024-02-18T12:00:20.000000Z"] → []  CLEARED
 - `exitid`: ["P1_1"] → []  CLEARED
 - Exit data appears ONLY in frame 20, not in frame 21+
-

SCENARIO 6: Multiple People (3 People Present)

Frame 50 Output:

json

```
{
  "camid": 1,
  "org_id": 100,
  "userid": 42,
  "Frame_Id": "FR_1_1708268450000",
  "Time_stamp": "2024-02-18T12:00:50.000000Z",
  "Frame_Count": 50,
  "Total_people_detected": 3,
  "Current_occupancy": 3,
  "People_ids": ["P1_2", "P1_3", "P1_4"],
  "Entry_time": [
    "2024-02-18T12:00:30.000000Z",
    "2024-02-18T12:00:35.000000Z",
    "2024-02-18T12:00:45.000000Z"
  ],
  "Exit_time": [],
  "exitid": [],
  "People_dwell_time": ["00:00:20", "00:00:15", "00:00:05"],
  "Confidence_scores": [0.8934, 0.9123, 0.8756],
  "Bounding_boxes": [
    [100, 150, 250, 450],
    [300, 140, 450, 460],
    [500, 160, 650, 440]
  ],
  "x": [175.0, 375.0, 575.0],
  "y": [300.0, 300.0, 300.0],
  "w": [150.0, 150.0, 150.0],
  "h": [300.0, 320.0, 280.0],
  "accuracy": [0.893, 0.912, 0.876],
```

```
"Total_entries": 4,  
"Total_exits": 1,  
"Net_count": 3,  
"Occupancy_percentage": 30.0,  
"Over_capacity_count": 0,  
"Average_dwell_time": "00:00:13",  
"Max_occupancy": 10,  
"Status": "",  
"is_alert_triggered": false,  
"Processing_Status": 1,  
"annotated_frame": null  
}
```

Key Points:

- All arrays have 3 elements (one per person)
 - Array indices correspond: People_ids[0] matches Entry_time[0], People_dwell_time[0], etc.
 - Average_dwell_time: $(20 + 15 + 5) / 3 = 13.33 \rightarrow "00:00:13"$
-

SCENARIO 7: One Person Exits (Out of 3)

Frame 55 Output:

json

```
{
  "camid": 1,
  "org_id": 100,
  "userid": 42,
  "Frame_Id": "FR_1_1708268455000",
  "Time_stamp": "2024-02-18T12:00:55.000000Z",
  "Frame_Count": 55,
  "Total_people_detected": 2,
  "Current_occupancy": 2,
  "People_ids": ["P1_2", "P1_4"],
  "Entry_time": [
    "2024-02-18T12:00:30.000000Z",
    "2024-02-18T12:00:45.000000Z"
  ],
  "Exit_time": ["2024-02-18T12:00:55.000000Z"],
  "exitid": ["P1_3"],
  "People_dwell_time": ["00:00:25", "00:00:10"],
  "Confidence_scores": [0.8987, 0.8812],
  "Bounding_boxes": [
    [102, 152, 252, 452],
    [502, 162, 652, 442]
  ],
  "x": [177.0, 577.0],
  "y": [302.0, 302.0],
  "w": [150.0, 150.0],
  "h": [300.0, 280.0],
  "accuracy": [0.899, 0.881],
  "Total_entries": 4,
  "Total_exits": 2,
```

```
"Net_count": 2,  
"Occupancy_percentage": 20.0,  
"Over_capacity_count": 0,  
"Average_dwell_time": "00:00:17",  
"Max_occupancy": 10,  
"Status": "",  
"is_alert_triggered": false,  
"Processing_Status": 1,  
"annotated_frame": null  
}
```

Key Points:

- `Total_people_detected`: $3 \rightarrow 2$
 - `People_ids`: P1_3 removed (middle person exited)
 - `Exit_time`: contains 1 exit timestamp
 - `exitid`: ["P1_3"]
 - All arrays now have 2 elements (2 remaining people)
 - `Total_exits`: $1 \rightarrow 2$
-

SCENARIO 8: Alert Triggered (80% Threshold)

Setup:

- Max_occupancy: 10
- alert_rate: 80
- Alert triggers at: $\text{round}(10 * 0.80) = 8$ people

Frame 100 Output (8 people enter):

json

```
{
  "camid": 1,
  "org_id": 100,
  "userid": 42,
  "Frame_Id": "FR_1_1708268500000",
  "Time_stamp": "2024-02-18T12:01:40.000000Z",
  "Frame_Count": 100,
  "Total_people_detected": 8,
  "Current_occupancy": 8,
  "People_ids": ["P1_5", "P1_6", "P1_7", "P1_8", "P1_9", "P1_10", "P1_11", "P1_12"],
  "Entry_time": [
    "2024-02-18T12:01:00.000000Z",
    "2024-02-18T12:01:05.000000Z",
    "2024-02-18T12:01:10.000000Z",
    "2024-02-18T12:01:15.000000Z",
    "2024-02-18T12:01:20.000000Z",
    "2024-02-18T12:01:25.000000Z",
    "2024-02-18T12:01:30.000000Z",
    "2024-02-18T12:01:35.000000Z"
  ],
  "Exit_time": [],
  "exitid": [],
  "People_dwell_time": ["00:00:40", "00:00:35", "00:00:30", "00:00:25", "00:00:20", "00:00:15", "00:00:10", "00:00:05"],
  "Confidence_scores": [0.89, 0.91, 0.87, 0.88, 0.90, 0.86, 0.85, 0.92],
  "Bounding_boxes": [
    [100, 150, 250, 450],
    [260, 140, 410, 460],
    [420, 160, 570, 440],
    [580, 155, 730, 455],
    [100, 500, 250, 800],
    [260, 490, 410, 810],
    [420, 510, 570, 790],
    [580, 505, 730, 805]
  ]
}
```

```

[110, 480, 260, 780],
[270, 490, 420, 790],
[430, 470, 580, 770],
[590, 485, 740, 785]
],
"x": [175.0, 335.0, 495.0, 655.0, 185.0, 345.0, 505.0, 665.0],
"y": [300.0, 300.0, 300.0, 305.0, 630.0, 640.0, 620.0, 635.0],
"w": [150.0, 150.0, 150.0, 150.0, 150.0, 150.0, 150.0, 150.0],
"h": [300.0, 320.0, 280.0, 300.0, 300.0, 300.0, 300.0, 300.0],
"accuracy": [0.890, 0.910, 0.870, 0.880, 0.900, 0.860, 0.850, 0.920],
"Total_entries": 12,
"Total_exits": 4,
"Net_count": 8,
"Occupancy_percentage": 80.0,
"Over_capacity_count": 0,
"Average_dwell_time": "00:00:22",
"Max_occupancy": 10,
"Status": "High Occupancy",
"is_alert_triggered": true,
"Processing_Status": 1,
"annotated_frame": null
}

```

Key Points:

- **Occupancy_percentage**: 80.0% (at threshold)
- **Status**: "High Occupancy"  ALERT TRIGGERED
- **is_alert_triggered**: true  NEW ALERT
- 8 people, 8 elements in all arrays

SCENARIO 9: Alert Suppressed (Same Frame + 1)

Frame 101 Output (Still 8 people):

json

```
{
  "camid": 1,
  "org_id": 100,
  "userid": 42,
  "Frame_Id": "FR_1_1708268501000",
  "Time_stamp": "2024-02-18T12:01:41.000000Z",
  "Frame_Count": 101,
  "Total_people_detected": 8,
  "Current_occupancy": 8,
  "People_ids": ["P1_5", "P1_6", "P1_7", "P1_8", "P1_9", "P1_10", "P1_11", "P1_12"],
  "Entry_time": [
    "2024-02-18T12:01:00.000000Z",
    "2024-02-18T12:01:05.000000Z",
    "2024-02-18T12:01:10.000000Z",
    "2024-02-18T12:01:15.000000Z",
    "2024-02-18T12:01:20.000000Z",
    "2024-02-18T12:01:25.000000Z",
    "2024-02-18T12:01:30.000000Z",
    "2024-02-18T12:01:35.000000Z"
  ],
  "Exit_time": [],
  "exitid": [],
  "People_dwell_time": ["00:00:41", "00:00:36", "00:00:31", "00:00:26", "00:00:21", "00:00:16", "00:00:11", "00:00:06"],
  "Confidence_scores": [0.89, 0.91, 0.88, 0.89, 0.90, 0.86, 0.85, 0.91],
  "Bounding_boxes": [
    [101, 151, 251, 451],
    [261, 141, 411, 461],
    [421, 161, 571, 441],
    [581, 156, 731, 456],
  ]
}
```

```

[111, 481, 261, 781],
[271, 491, 421, 791],
[431, 471, 581, 771],
[591, 486, 741, 786]
],
"x": [176.0, 336.0, 496.0, 656.0, 186.0, 346.0, 506.0, 666.0],
"y": [301.0, 301.0, 301.0, 306.0, 631.0, 641.0, 621.0, 636.0],
"w": [150.0, 150.0, 150.0, 150.0, 150.0, 150.0, 150.0, 150.0],
"h": [300.0, 320.0, 280.0, 300.0, 300.0, 300.0, 300.0, 300.0],
"accuracy": [0.890, 0.910, 0.880, 0.890, 0.900, 0.860, 0.850, 0.910],
"Total_entries": 12,
"Total_exits": 4,
"Net_count": 8,
"Occupancy_percentage": 80.0,
"Over_capacity_count": 0,
"Average_dwell_time": "00:00:23",
"Max_occupancy": 10,
"Status": "",
"is_alert_triggered": false,
"Processing_Status": 1,
"annotated_frame": null
}

```

Key Points:

- Still 8 people (same as frame 100)
- `Status`: ""  SUPPRESSED (was "High Occupancy" in frame 100)
- `is_alert_triggered`: false  SUPPRESSED
- Prevents alert spam - alert sent only once

SCENARIO 10: Over Capacity

Frame 150 Output (15 people, max=10):

json

```
{
  "camid": 1,
  "org_id": 100,
  "userid": 42,
  "Frame_Id": "FR_1_1708268550000",
  "Time_stamp": "2024-02-18T12:02:30.000000Z",
  "Frame_Count": 150,
  "Total_people_detected": 15,
  "Current_occupancy": 15,
  "People_ids": ["P1_5", "P1_6", "P1_7", "P1_8", "P1_9", "P1_10", "P1_11", "P1_12"],
  "Entry_time": [
    "2024-02-18T12:01:00.000000Z",
    "2024-02-18T12:01:05.000000Z",
    "2024-02-18T12:01:10.000000Z",
    "2024-02-18T12:01:15.000000Z",
    "2024-02-18T12:01:20.000000Z",
    "2024-02-18T12:01:25.000000Z",
    "2024-02-18T12:01:30.000000Z",
    "2024-02-18T12:01:35.000000Z",
    "2024-02-18T12:01:45.000000Z",
    "2024-02-18T12:01:55.000000Z",
    "2024-02-18T12:02:05.000000Z",
    "2024-02-18T12:02:10.000000Z",
    "2024-02-18T12:02:15.000000Z",
    "2024-02-18T12:02:20.000000Z",
    "2024-02-18T12:02:25.000000Z"
  ],
  "Exit_time": [],
  "exitid": [],
```

```

"Personnel_dwell_time": ["00:01:30", "00:01:25", "00:01:20", "00:01:15", "00:01:10",
"Confidence_scores": [0.89, 0.91, 0.87, 0.88, 0.90, 0.86, 0.85, 0.92, 0.88, 0.87, 0.91,
"Bounding_boxes": [
[100, 150, 250, 450],
[260, 140, 410, 460],
[420, 160, 570, 440],
[580, 155, 730, 455],
[110, 480, 260, 780],
[270, 490, 420, 790],
[430, 470, 580, 770],
[590, 485, 740, 785],
[750, 150, 900, 450],
[910, 140, 1060, 460],
[1070, 160, 1220, 440],
[105, 810, 255, 1110],
[265, 820, 415, 1120],
[425, 800, 575, 1100],
[585, 815, 735, 1115]
],
"x": [175.0, 335.0, 495.0, 655.0, 185.0, 345.0, 505.0, 665.0, 825.0, 985.0, 1145.0, 18
"y": [300.0, 300.0, 300.0, 305.0, 630.0, 640.0, 620.0, 635.0, 300.0, 300.0, 300.0, 960
"w": [150.0, 150.0, 150.0, 150.0, 150.0, 150.0, 150.0, 150.0, 150.0, 150.0, 150.0, 150
"h": [300.0, 320.0, 280.0, 300.0, 300.0, 300.0, 300.0, 300.0, 300.0, 320.0, 280.0, 300
"accuracy": [0.890, 0.910, 0.870, 0.880, 0.900, 0.860, 0.850, 0.920, 0.880, 0.870, 0.9
"Total_entries": 19,
"Total_exits": 4,
"Net_count": 15,
"Occupancy_percentage": 150.0,
"Over_capacity_count": 5,
"Average_dwell_time": "00:00:51",
"Max_occupancy": 10,

```



```
"Status": "",  
"is_alert_triggered": false,  
"Processing_Status": 1,  
"annotated_frame": null  
}
```

Key Points:

- 15 people (5 over capacity)
 - `Occupancy_percentage`: 150.0% (NOT capped at 100%)
 - `Over_capacity_count`: 5 (15 - 10)
 - `Status`: "" (alert already triggered earlier, now suppressed)
 - All arrays have 15 elements
-

SCENARIO 11: Error Response

Output When Model Fails:

json

```
{  
  "camid": 1,  
  "org_id": 100,  
  "userid": 42,  
  "Frame_Id": "ERROR_1708268600000",  
  "Time_stamp": "2024-02-18T12:03:20.000000Z",  
  "Frame_Count": 200,  
  "Total_people_detected": 0,  
  "Current_occupancy": 0,  
  "People_ids": [],  
  "Entry_time": [],  
  "Exit_time": [],  
  "exitid": [],  
  "People_dwell_time": [],  
  "Confidence_scores": [],  
  "Bounding_boxes": [],  
  "x": [],  
  "y": [],  
  "w": [],  
  "h": [],  
  "accuracy": [],  
  "Total_entries": 19,  
  "Total_exits": 4,  
  "Net_count": 15,  
  "Occupancy_percentage": 0.0,  
  "Over_capacity_count": 0,  
  "Average_dwell_time": "00:00:00",  
  "Max_occupancy": 10,  
  "Status": "Error",  
}
```

```
"is_alert_triggered": false,  
"Processing_Status": 0,  
"annotated_frame": null,  
"error_message": "YOLO detection failed: CUDA out of memory"  
}
```

Key Points:

- `Processing_Status`: 0 (error)
 - `Status`: "Error"
 - `error_message`: Added (only in error responses)
 - All arrays empty
 - Counters retain last valid values
-

SUMMARY OF KEY PATTERNS

Empty Arrays Pattern:

json

```
{  
  "People_ids": [],  
  "Entry_time": [],  
  "Exit_time": [],  
  "exitid": [],  
  "People_dwell_time": [],  
  "Confidence_scores": [],  
  "Bounding_boxes": [],  
  "x": [],  
  "y": [],  
  "w": [],  
  "h": [],  
  "accuracy": []  
}
```

Single Person Pattern:

json

```
{  
  "People_ids": ["P1_1"],  
  "Entry_time": ["2024-02-18T12:00:10.000Z"],  
  "People_dwell_time": ["00:00:05"],  
  "Confidence_scores": [0.893],  
  "Bounding_boxes": [[245, 180, 395, 480]],  
  "x": [320.0],  
  "y": [330.0],  
  "w": [150.0],  
  "h": [300.0],  
  "accuracy": [0.893]  
}
```

Exit Pattern (Only in exit frame):

json

```
{  
  "Exit_time": ["2024-02-18T12:00:20.000Z"],  
  "exitid": ["P1_1"]  
}
```

Alert Triggered Pattern:

json

```
{  
  "Status": "High Occupancy",  
  "is_alert_triggered": true  
}
```

Alert Suppressed Pattern:

json

```
{  
  "Status": "",  
  "is_alert_triggered": false  
}
```